

BETA TEST PLAN

1. Project context

HuRI is a research-driven platform that has the objective to act as a universal middleware for speech and movement for any various physical or digital human embodiments. The focus on our part is to have embodiments as human as possible, regarding the emotion aspect, and try to go beyond the uncanny valley. The goal is to create a modular ecosystem where any combination of AI can control any robot or any virtual avatar. The project focuses on a highly stable architecture, scalable cloud deployment, and innovative emotional expression through motion and voice.

2. User role

Role Name	Description
Module Developer	Uses the library to build and package custom plugins or specialized modules for specific use cases.
DevOps Engineer	Handles the containerization, orchestration, and distribution of the library across different network nodes or edge devices.
Client	Integrates the library into their own projects without concern for its internal implementation.

3. Feature table

Feature ID	User role	Feature name	Short description
F1	Module Developer	Create a Module	Build and integrate modules into the library.
F2	Module Developer & DevOps Engineer	Configure HuRI	Define available modules and deployment settings for HuRI.
F3	Client	Configure Modules	Define and load different module combinations through a configuration file.

F4	Everyone	Run in Parallel	Distribute computation across 1 to N machines to balance the payload.
F5	Everyone	Handle Multi-client	Handle 1 to N simultaneous client instances with separated discussions.
F6	Everyone	Detect Voice Activity	Detect when the user is speaking.
F7	Everyone	Transcribe Speech	Generate text from an audio speech.
F8	Everyone	Generate Speech	Generate speech audio from a given text input.
F9	Everyone	Generate Body Movement	Generate body movements by placing points in space.
F10	Everyone	Retrieve and Augment	Retrieve text from files, saved texts, or past conversations and generate an output.
F11	Everyone	Recognise Speech Emotion	Recognise emotions from user speech.
F12	Everyone	Link Emotion to Input	Link a transcript with its associated emotion.
F13	Everyone	Manage Artificial Memory	Store and manage an artificial memory for HuRI.
F14	Everyone	Send User Audio	Transmit user audio to the HuRI system.
F15	Everyone	Send User Text	Transmit user text to the HuRI system.
F16	Everyone	Receive Generated Audio	Deliver generated audio back to the user.
F17	Everyone	Receive Generated Text	Deliver generated text back to the user.

F18	Everyone	Store Vectorised Data	Store and retrieve vectorised data through a dedicated service.
F19	Everyone	Call LLM Service	Generate a response by calling an external LLM service.

4. Success Criteria

Feature ID	Key success criteria	Indicator/metric	Result
F1	Verify modules can be written and run inside HuRI.	20 attempts -- expected 100%	Achieved (/20)
F2	Check that available modules and cluster settings correspond to the configuration.	5 configurations -- expected 100%	Achieved (/5)
F3	Confirm that a specific module combination loads from a config file.	5 files -- expected 100%	Achieved (/5)
F4	Test that HuRI runs across multiple machines with computation distributed.	1 machine / N machines	Scenario achieved (/2)
F5	Verify multiple clients stay isolated with separate discussions.	3 simultaneous instances -- expected 100% isolation	Instances isolated (/3)
F6	Check that speaking and silence are correctly detected.	10 voice samples -- expected 100%	Detections successful (/10)
F7	Confirm spoken input is transcribed accurately.	20 spoken phrases -- expected 80% accuracy	Phrases correct (/20)

F8	Verify a speech audio file with human characteristic is generated from a text input accurately.	10 text inputs -- expected 100% generation -- expected 60% human feeling*	Files generated (/10) -- Human feedback in %
F9	Assess whether body movements feel human-like.	10 movement sequences -- expected 60% human feeling	Human feedback in %
F10	Check that relevant saved content is retrieved to build an answer.	10 queries on saved text -- expected 100% retrieval	Relevant text found (/10)
F11	Verify the emotion in user speech is correctly identified.	10 emotion samples -- expected 60% accuracy	Emotions recognised (/10)
F12	Confirm the detected emotion is associated with the transcript.	10 emotional contexts -- expected 60%	Contexts linked (/10)
F13	Test that information is retained, updated, and forgotten over time.	10 pieces of info over time -- expected 60%	Prompts saved and treated (/10)
F14	Verify user audio is transmitted correctly to the HuRI system.	10 audio inputs -- expected 100%	Inputs received (/10)
F15	Confirm user text is transmitted correctly to the HuRI system.	10 text inputs -- expected 100%	Inputs received (/10)
F16	Check that generated audio is correctly delivered back to the user.	10 audio files -- expected 100%	Files heard (/10)
F17	Confirm generated text is correctly delivered back to the user.	10 text outputs -- expected 100%	Outputs displayed (/10)

F18	Verify vectorised data is stored and retrieved accurately.	10 operations -- expected 100%	Operations successful (/10)
F19	Test that a coherent response is returned from the external LLM service.	10 prompts -- expected 100% generation	Answers generated (/10)

*Experiments will be conducted on several people and a Godspeed-based questionnaire will evaluate the human feeling